Status	Finished
Started	Tuesday, 20 October 2026, 4:43 PM
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Duration	8 mins
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. There must be a mechanism to access middle element directly

d. Must use a sorted array

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓

b. Linear search is more efficient than binary search on large lists.

c. Linear search requires the list to be sorted.

d. Linear search divides the list into halves.

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element

b. By dividing the list length by two ✓

c. By comparing each element sequentially

d. By using a hash function

Question **4**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Sequential search
c. Hash search
d. Binary search ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is large and unsorted
c. When the list is small and sorted
d. When the list is small and unsorted ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)
b. Binary search
c. Hash search
d. Linear search ✓

Question **7**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000
b. 10 ✓
c. 100
d. 500

Question **8**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. In the right sublist ✓
c. In the middle of the list
d. At the beginning of the list

Question **9**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The search stops
b. The middle element is chosen randomly
c. The higher middle element is chosen as the middle element
d. The lower middle element is chosen as the middle element ✗

Question **10**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Insertion

c. Selection

d. Bubble ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 99 ✓
- b. 90 and 100
- c. 89 and 94
- d. 94 and 99

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Sorting the list before searching
- b. Checking each element sequentially ✓
- c. Dividing the list in half repeatedly
- d. Skipping every second element

Question **13**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search stops immediately ✓
- b. The search backtracks to find duplicate elements
- c. The search starts over from the beginning
- d. The search continues until the end of the list

Question **14**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(log n)
- b. O(1)
- c. O(n log n)
- d. O🥐 ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(log n) ✓
c. O👎
d. O(n log n)